

Global Corporate ESG & Decarbonization Strategy Dashboard

1. Document Overview

- **Objective:** To establish an end-to-end data transformation and monitoring application that calculates and consolidates corporate Scope 1, Scope 2, and Scope 3 greenhouse gas (GHG) emissions across international production sites.
- **Target Audience:** Chief Sustainability Officer (CSO), Operations Directors, and ESG Compliance Auditors.

2. Business Objective & Problem Statement

Global regulatory frameworks demand transparent, auditable carbon accounting. Currently, consumption data (MWh, fuel volumes) and logistics metrics (ton-kilometres) are scattered across regional operations. Greenhouse gas calculations are done manually in un-auditable Excel files, increasing compliance risks and masking operational inefficiencies that prevent the firm from meeting its 2030 decarbonization targets.

3. Business Use Cases & Analytic Scenarios

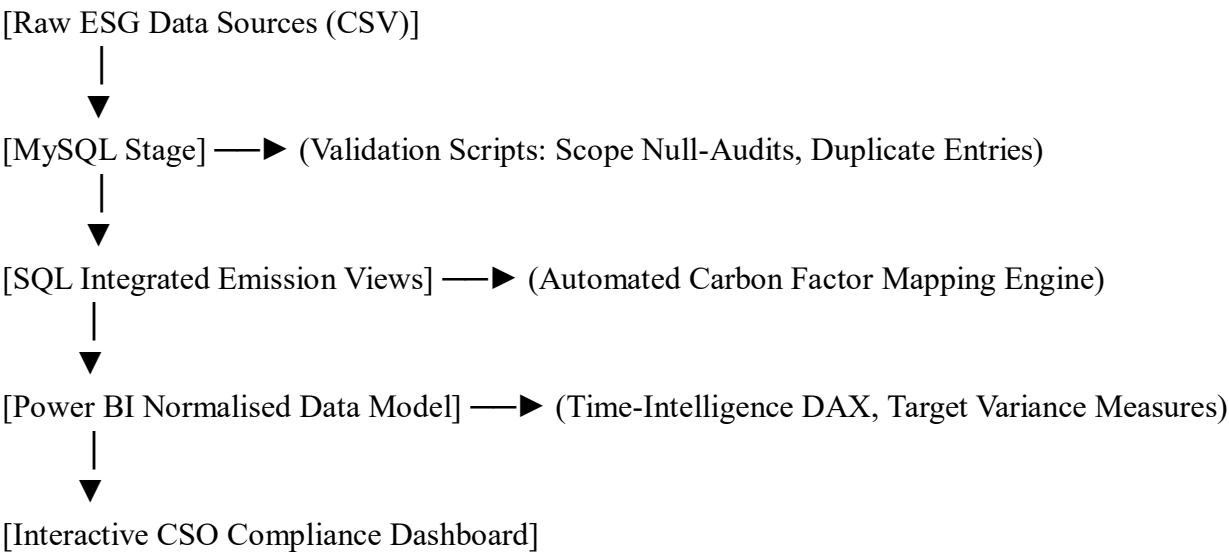
Scenario A: Regulatory Audit Readiness (The ESG Compliance Auditor Perspective)

- **Goal:** Trace every single carbon metric back to its original raw consumption entry and time-varying emission factor.
- **Business Logic:** Provide structural data lineage where emissions are calculated dynamically by multi-table associations based on the explicit year of activity.

Scenario B: Decarbonization Trajectory Over Time (The CSO Perspective)

- **Goal:** Track year-over-year (YoY) emission intensity reduction against corporate commitments.
- **Business Logic:** If an industrial plant's annual emission reduction variance falls behind its stated reduction target, flag it visually as "At Risk."

4. Technical & Functional Architecture



5. Data Requirements & Source Mapping

Entity / Table	Source Grain	Core Attributes Required	Key Business Validation Rules
esg_facilities_dim	Per Facility	Plant_ID, Name, Country, Segment, Target	Baseline asset catalogue; must serve as a strict validation lookup.
esg_emissions_factors_dim	Annual, by Activity	Year, Activity_Type, Scope, Emission_Factor	Must account for changing grid factors across 2024, 2025, and 2026.
esg_energy_consumption_fact	Monthly, by Plant	Reporting_Date, Plant_ID, Energy_Source, Quantity_MWh	Identify and deduplicate multi-reported records within a single month.
esg_logistics_fact	Per Shipment	Shipment_ID, Plant_ID, Shipment_Date, Mode_of_Transport, Weight_Tons, Distance_KM	Distance and weight must be non-zero positive values.

6. Key Performance Indicators (KPIs) & Metrics

- **Scope 1 / Scope 2 Emissions (Metric Tons CO₂e):**
Emissions = Quantity (MWh) × Emission Factor
- **Scope 3 Supply Chain Emissions (Metric Tons CO₂e):**
Scope 3 Emissions = Weight (Tons) × Distance (KM) × Transport Emission Factor
- **Carbon Target Variance (%):**
$$\text{Variance} = \frac{\text{Actual Emissions Reduction}}{\text{Target Emissions Reduction}} - 1$$

7. Reporting & Data Visualisation Scope

- **CSO Control Tower View:** Global metric cards showcasing total carbon footprint broken down cleanly into Scope 1, Scope 2, and Scope 3 emissions.
- **Regional Performance Decomposition Tree:** An interactive data structure allowing executives to break down absolute emissions by Scope → Region → Plant → Fuel Source.
- **Decarbonization Trajectory Visual:** Linear comparison chart showing actual emissions lines alongside targeted reduction pathways up to 2026.